

SOAL UTS STATISTIKA MATEMATIKA 1

1. In *USA Today* (Sept. 5, 1996), the results of a survey involving the use of sleepwear while traveling were listed as follows:

	Male	Female	Total
Underwear	0.220	0.024	0.244
Nightgown	0.002	0.180	0.182
Nothing	0.160	0.018	0.178
Pajamas	0.102	0.073	0.175
T-shirt	0.046	0.088	0.134
Other	0.084	0.003	0.087

- (a) What is the probability that a traveler is a female who sleeps in the nude?
- (b) What is the probability that a traveler is male?
- (c) Assuming the traveler is male, what is the probability that he sleeps in pajamas?
- (d) What is the probability that a traveler is male if the traveler sleeps in pajamas or a T-shirt?
2. A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.75. Of those that purchase latex paint, 60% also purchase rollers. But only 30% of semigloss paint buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?
3. Suppose that the probabilities are 0.4, 0.3, 0.2, and 0.1, respectively, that 0, 1, 2, or 3 power failures will strike a certain subdivision in any given year.
- a. Show that the probabilities construct density probability function!
- b. Find mean and variance of that probabilities!